$FINA$ 6339 : $Lecture$ 11 & $Lecture$ 12.

[Portfolio Optimization (Part II)] [R].)

© Milivoje Davidovic.

Northeastern University, Finance Academic Group.

E-Mail: *m.davidovic@northeastern.edu*

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```
# FINA 6339: Lecture 9 & Lecture 10
# [Portfolio Optimization: Part II]
#########################################################



####################
# Data Description:
##################

## Daily simple and log returns for Nike, Amazon,
## Google, Tesla, and Microsoft, covering the period
# Jan 2020 - December 2022.

## The data set will be posted on Canvas as
## FiveStocksSimpleRet.csv and FiveStocksLogRet.csv file.

#########################################################

# Define the working directory
setwd("C:\\Users\\Milivoje Davidovic\\Desktop")

# Import data set into R environment
mydata = read.csv("FiveStocksLogRet.csv") # Log returns
attach(mydata)

# Create Date vector
Date=seq(as.Date("2020-01-01"), by=1, len=756)

# Plot Log Returns
windows(8,10)
plot(Date, NIKE, type="l", col="red",
     main="Blue Chips Log Returns (Jan 2020 - Dec 2022)",
     ylab="Return",ylim=c(-0.06,0.06))
lines(Date, AMZN, col="blue", lty=2)
lines(Date, GOOGLE, col="green", lty=2)
lines(Date, TESLA, col="orange", lty=2)
lines(Date, MSFT, col="black", lty=2)
```

```r
grid(nx=20,ny=20)
# Add a legend
legend("topleft", legend=c("NIKE", "AMZN","GOOGLE","TESLA","MSFT"),
       col=c("red", "blue","green","orange","black"),
       lty=1, cex=0.8)
```

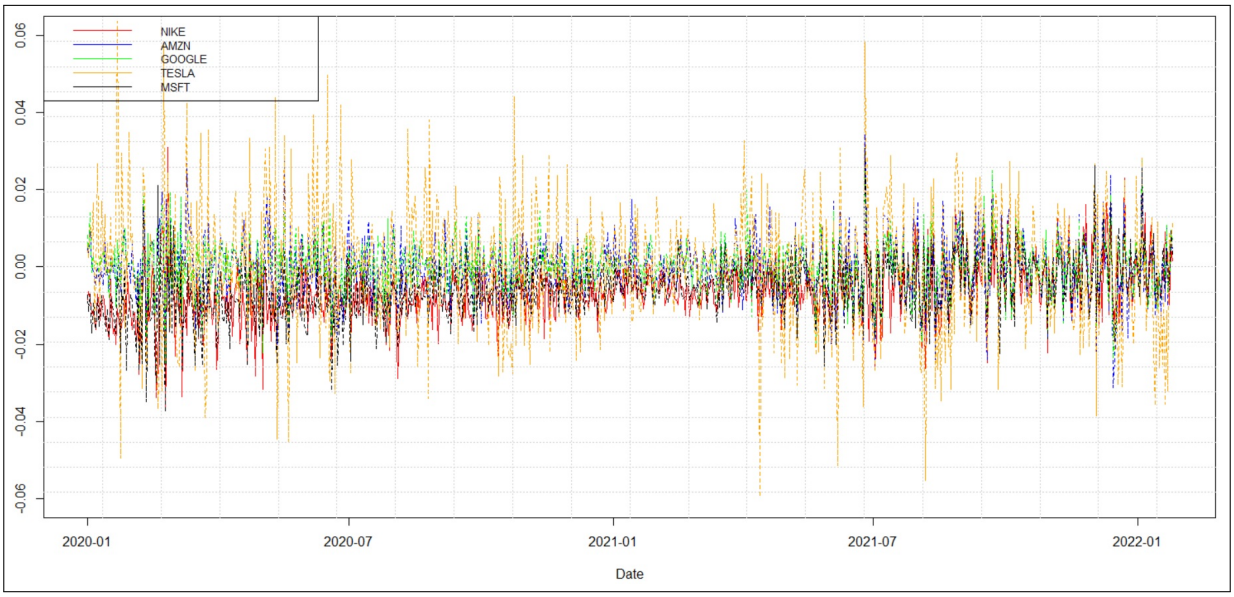

Figure 1: Log Returns (Blue Chips).

```r
#######################################
# Mean-Variance Optimization Framework.
#######################################

##################################################
# Portfolio Optimization (Variance Minimization).
##################################################
returns=ts(mydata[,2:6]) # to cut of the Date
# ts: create time series vector
print(head(returns, 5))
```

```
##              NIKE         AMZN       GOOGLE       TESLA         MSFT
## [1,] -0.008429582  0.005297210  0.008279190 0.005853294 -0.007248433
## [2,] -0.006309223  0.002431931  0.004104826 0.002467612 -0.011430121
## [3,] -0.008244359  0.009898444  0.013994444 0.010779879 -0.006894031
## [4,] -0.012141780  0.000537850 -0.001431445 0.007150800 -0.017021422
## [5,] -0.010986244 -0.001391101  0.003801931 0.016585289 -0.009093842
```

```r
# Get a character vector of the fund names
funds <- colnames(returns)

# Install/call the required libraries
suppressMessages(library(PortfolioAnalytics))
```

```r
suppressMessages(library(foreach))
suppressMessages(library(iterators))
```

```r
suppressMessages(library(ROI))
suppressMessages(library(ROI.plugin.quadprog))
suppressMessages(library(ROI.plugin.glpk))

# Create portfolio object
MinVar <- portfolio.spec(assets=funds)

# Add full investment constraint to the portfolio object
MinVar <- add.constraint(portfolio=MinVar, type="full_investment")

# Add objective to minimize variance
MinVar <- add.objective(portfolio=MinVar, type="risk", name="var")

# Run the optimization procedure (Global Min variance)
opt_gmv <- optimize.portfolio(R=returns, portfolio=MinVar,
                              optimize_method="ROI", trace=TRUE)

# OPTIMAL WEIGHTS
print(opt_gmv)
```

```
## ***********************************
## PortfolioAnalytics Optimization
## ***********************************
##
## Call:
## optimize.portfolio(R = returns, portfolio = MinVar, optimize_method = "ROI",
##     trace = TRUE)
##
## Optimal Weights:
##    NIKE    AMZN  GOOGLE   TESLA    MSFT
##  0.3445  0.0989  0.4917 -0.0214  0.0863
##
## Objective Measure:
##   StdDev
## 0.006303
```

```r
# Objective: StdDev = 0.006303

##########################################
# Visualization of the Optimal portfolio.
##########################################
windows(8,10)
plot(opt_gmv, chart.assets=TRUE)
```

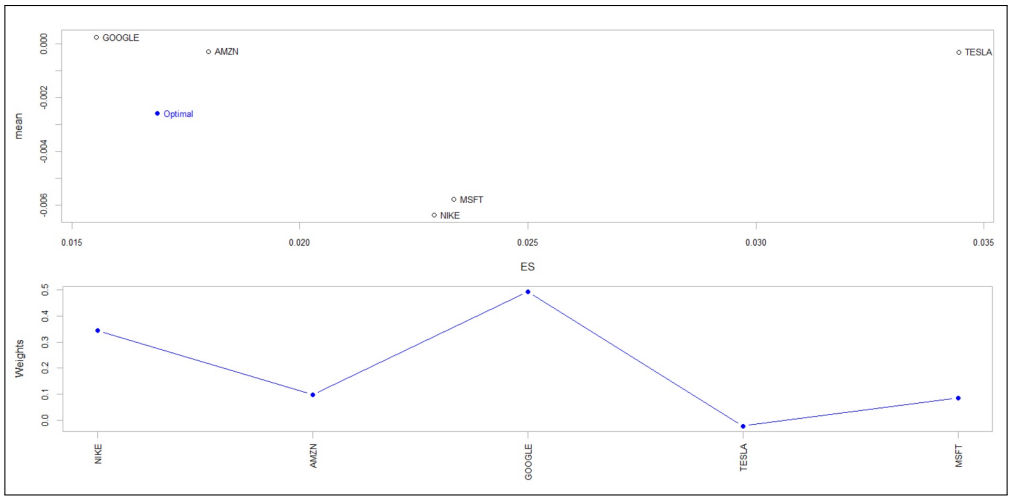

Figure 2: Optimal Portfolio Weights (Min-Var).

```
############################
# Min-Var Optimal Portfolio.
############################

# Create the optimal portfolio using these weights
returns=mydata[,2:6]
OptPort1 = (0.3445*returns$NIKE) + (0.0989*returns$AMZN)+
  (0.4917*returns$GOOGLE)+(-0.0214*returns$TESLA)+
  (0.0863*returns$MSFT)

# Distribution of your optimal portfolio
windows(8,10)
hist(OptPort1, main="Optimal Min-Var Portfolio Retruns",
     xlab="Returns",col="orange",breaks = 30)
```

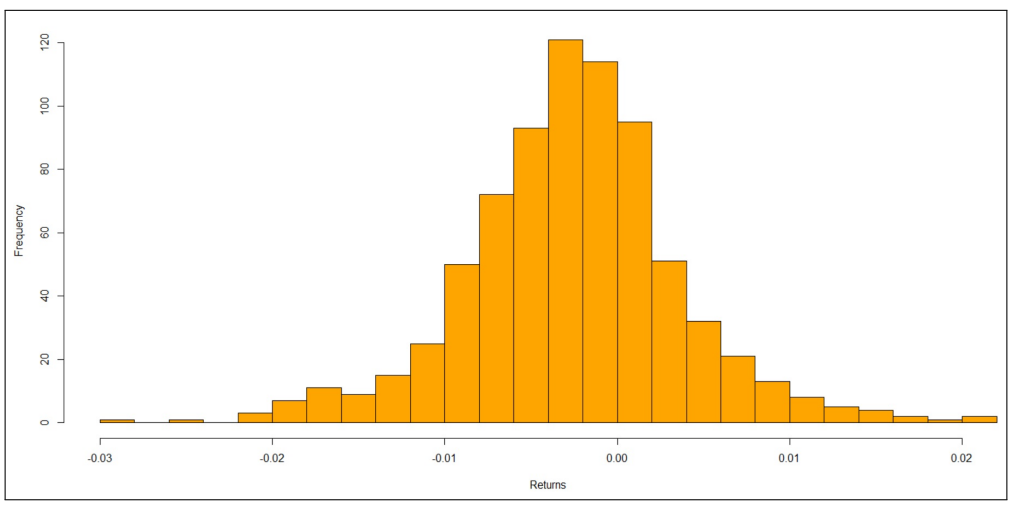

Figure 3: Optimal Min-Var Portfolio Returns.

```r
##########################################
# Maximizing Mean Return (Liner Utility)
##########################################

# Create portfolio object
MaxRet <- portfolio.spec(assets=funds)

# Add constraints to the portfolio object

## Full investment constraint
MaxRet <- add.constraint(portfolio=MaxRet,
                         type="full_investment")

## Add objective to the portfolio object
MaxRet <- add.objective(portfolio=MaxRet,
                        type="return", name = "mean")

## Note: Box Constraint is necessary for MaxRet problem
## Otherwise, the weights would go from -inf to inf
MaxRet <- add.constraint(portfolio=MaxRet, type="box",
                         min=c(0, 0, 0, 0,0),
                         max=c(0.25,0.25,0.25,0.25,0.25))

## To see the summary of your portfolio specification
print(MaxRet)
```

```
## ***************************************************
## PortfolioAnalytics Portfolio Specification
## ***************************************************
##
## Call:
## portfolio.spec(assets = funds)
##
## Number of assets: 5
## Asset Names
## [1] "NIKE"   "AMZN"   "GOOGLE" "TESLA"  "MSFT"
##
## Constraints
## Enabled constraint types
##      - full_investment
##      - box
##
## Objectives:
## Enabled objective names
##      - mean
```

```r
# Run the optimization
returns=ts(mydata[,2:6]) # returns must be a ts object
opt_max_ret <- optimize.portfolio(R=returns, portfolio=MaxRet,
                                  optimize_method="ROI", trace=TRUE)

# Optimal Weights
print(opt_max_ret)
```

```
## **********************************
```

```
## PortfolioAnalytics Optimization
## **********************************
##
## Call:
## optimize.portfolio(R = returns, portfolio = MaxRet, optimize_method = "ROI",
##     trace = TRUE)
##
## Optimal Weights:
##   NIKE   AMZN GOOGLE  TESLA   MSFT
##   0.00   0.25   0.25   0.25   0.25
##
## Objective Measure:
##      mean
## -0.001539
```

```
##########################################
# Visualization of the Optimal portfolio.
##########################################
windows(8,10)
plot(opt_max_ret, chart.assets=TRUE)
```

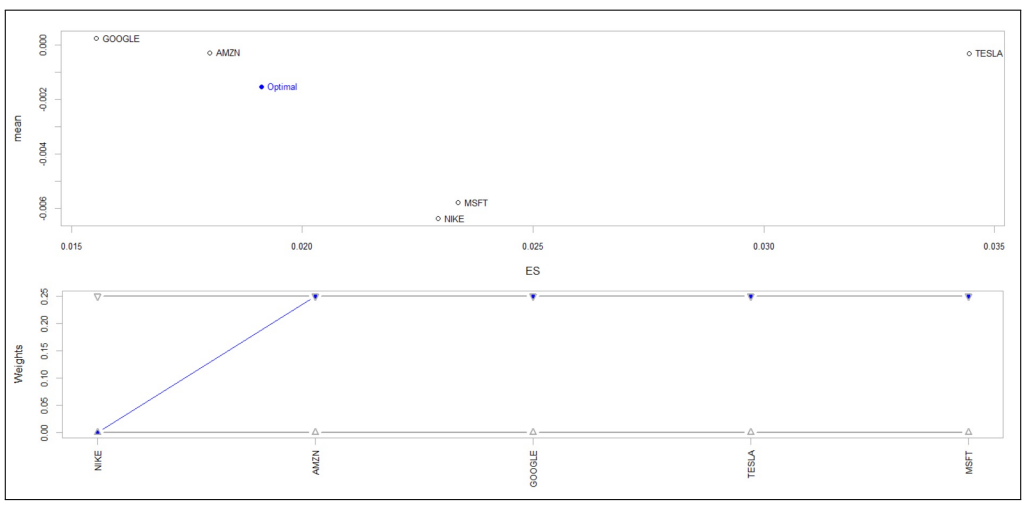

Figure 4: Optimal Portfolio (Max-Ret).

```
################################
# Max-Return Optimal Portfolio.
################################

# Create the optimal portfolio using these weights
returns=mydata[,2:6]
OptPort2 = (0.25*returns$NIKE) + (0.25*returns$AMZN)+
  (0.25*returns$GOOGLE)+(0.25*returns$TESLA)+
  (0.25*returns$MSFT)

# Distribution of your optimal portfolio
windows(8,10)
hist(OptPort2, main="Optimal Max-Ret Portfolio Returns",
     xlab="Returns",col="green",breaks = 30)
```

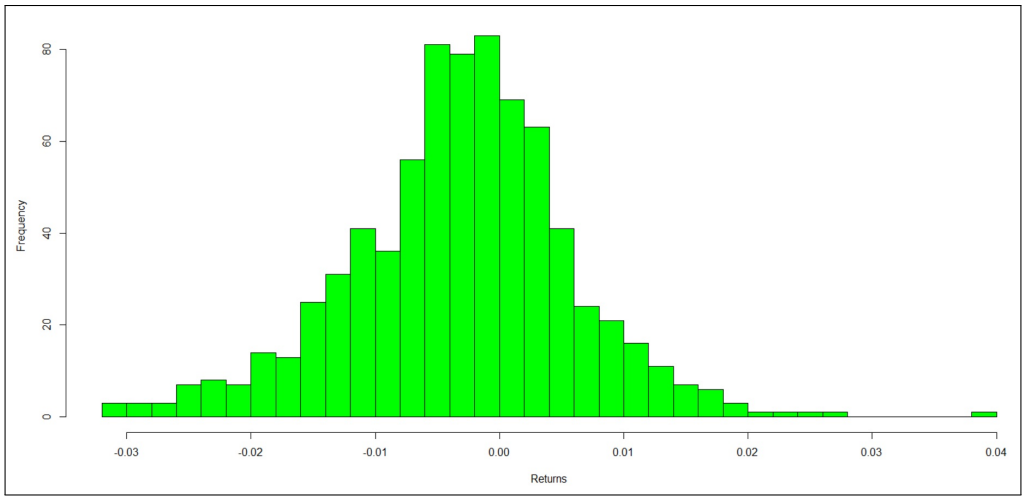

Figure 5: Optimal Max-Return Portfolio Returns.

```
#########################
# Risk Parity Portfolio.
#########################

# Example: Consider returns for these five companies;
# We want to use these log returns to to build a risk
# parity and tangency portfolios, respectively.

# Install/call required Packages
library(IDPmisc)
library(riskParityPortfolio)
library(fPortfolio)
```

```
## Loading required package: timeDate

##
## Attaching package: 'timeDate'

## The following objects are masked from 'package:PerformanceAnalytics':
##
##     kurtosis, skewness

## Loading required package: timeSeries

##
## Attaching package: 'timeSeries'

## The following object is masked from 'package:zoo':
##
##     time<-

## The following object is masked from 'mydata':
##
##     MSFT

## Loading required package: fBasics

##
## Attaching package: 'fBasics'
```

```
## The following object is masked from 'package:ROI':
##
##     vech

## The following objects are masked from 'package:PerformanceAnalytics':
##
##     kurtosis, skewness

## Loading required package: fAssets
```

```r
# Calculate covariance matrix of the returns
Sigma <- cov(returns)

# Compute risk parity portfolio
Parity_Port <- riskParityPortfolio(Sigma)
Parity_Port
```

```
## $w
##      NIKE      AMZN    GOOGLE     TESLA      MSFT
## 0.2353130 0.1980726 0.2300589 0.1273860 0.2091694
##
## $relative_risk_contribution
##   NIKE   AMZN GOOGLE  TESLA   MSFT
##    0.2    0.2    0.2    0.2    0.2
##
## $obj_fun
## [1] 1.632128
##
## $is_feasible
## [1] TRUE
```

```r
# Compute tangency portfolio (Max Sharpe ratio)
Tang_Port <- tangencyPortfolio(as.timeSeries(returns),
                               constraints = "LongOnly")
Tang_Port # Overconcentrated
```

```
##
## Title:
##  MV Tangency Portfolio
##  Estimator:         covEstimator
##  Solver:            solveRquadprog
##  Optimize:          minRisk
##  Constraints:       LongOnly
##
## Portfolio Weights:
##   NIKE   AMZN GOOGLE  TESLA   MSFT
##      0      0      1      0      0
##
## Covariance Risk Budgets:
##   NIKE   AMZN GOOGLE  TESLA   MSFT
##      0      0      1      0      0
##
## Target Returns and Risks:
##   mean    Cov   CVaR    VaR
## 0.0002 0.0070 0.0154 0.0117
##
```

```
## Description:
##  Sun Feb 18 13:49:40 2024 by user: Milivoje Davidovic
```

```r
# Extract Portfolio Weights for both Portfolios
weights <- rbind(Parity_Port$w, getWeights(Tang_Port))
row.names(weights)<-c("Parity Portfolio", "Tangency Portfolio")


# Plot these portfolio weights to compare the two portfolios
windows(8,10)
barplot(weights, main = "", xlab = "Stocks", ylab = "Dollars",
        beside = TRUE, legend = TRUE, col=c("blue", "orange"),
        args.legend = list(bg = "white"),
        ylim=c(0,1.1))
grid(nx=20,ny=20)
```

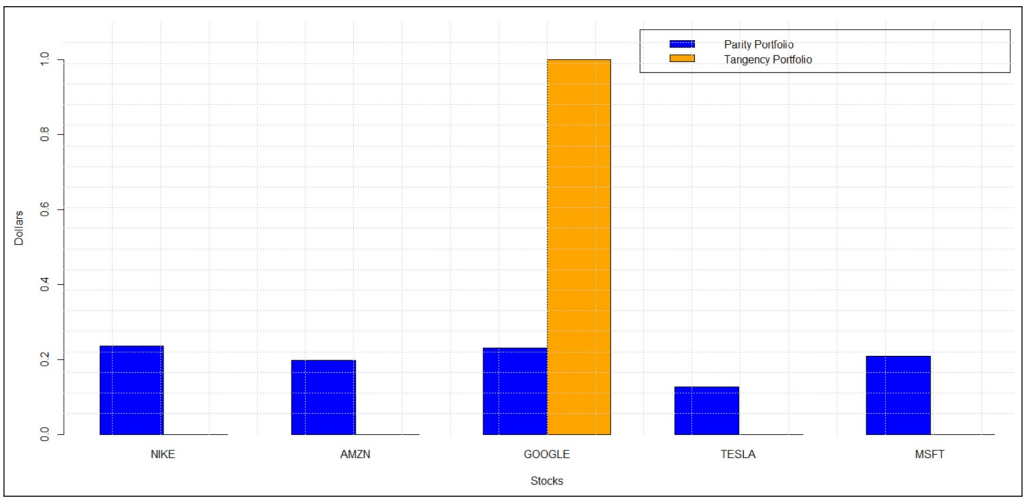

Figure 6: Risk-parity vs. Tangency Portfolio.

```r
# Create the optimal portfolio using risk-parity weights
returns=mydata[,2:6]
OptPort3 = (0.2353130*returns$NIKE) + (0.1980726*returns$AMZN)+
  (0.2300589*returns$GOOGLE)+(0.1273860*returns$TESLA)+
  (0.2091694*returns$MSFT)

# Distribution of your optimal portfolio
windows(8,10)
hist(OptPort3, main="Optimal Risk-Parity Portfolio Returns",
     xlab="Returns",col="blue",breaks = 30)
```

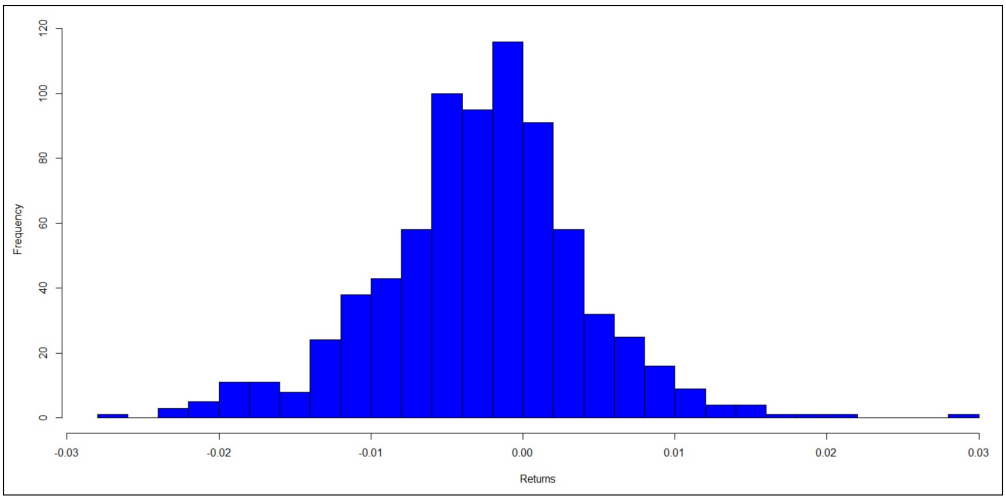

Figure 7: Optimal Risk-parity Portfolio Returns.

```
#####################################################
# Compare Optimal Solutions
# [Risk-parity vs. Min-Var vs. Max-Ret Allocations]
#####################################################


data <- data.frame(Weights = c(0.3445, 0.0989, 0.4917, -0.0214, 0.0863,
                               0.00,0.25,0.25,0.25,0.25,
                               0.2353,0.1981,0.2301,0.1274,0.2092),  # Create example data
                   Optimal_Portfolios = rep(c("Min-Var",
                                              "Max-Ret",
                                              "Risk-Parity"),
                                            each = 5),
                   Companies = c("NIKE",
                                 "AMZN",
                                 "GOOGLE",
                                 "TESLA","MSFT"))
# Check you weights distribution
data
```

```
##    Weights Optimal_Portfolios Companies
## 1   0.3445            Min-Var      NIKE
## 2   0.0989            Min-Var      AMZN
## 3   0.4917            Min-Var    GOOGLE
## 4  -0.0214            Min-Var     TESLA
## 5   0.0863            Min-Var      MSFT
## 6   0.0000            Max-Ret      NIKE
## 7   0.2500            Max-Ret      AMZN
## 8   0.2500            Max-Ret    GOOGLE
## 9   0.2500            Max-Ret     TESLA
## 10  0.2500            Max-Ret      MSFT
## 11  0.2353        Risk-Parity      NIKE
## 12  0.1981        Risk-Parity      AMZN
## 13  0.2301        Risk-Parity    GOOGLE
## 14  0.1274        Risk-Parity     TESLA
## 15  0.2092        Risk-Parity      MSFT
```

```r
# Plot using ggplot (Grouped barplot)
library(ggplot2)
windows(8,10)
ggplot(data,
       aes(x = Optimal_Portfolios,
           y = Weights,
           fill = Companies)) +
  geom_bar(stat = "identity",
           position = "dodge")+
  geom_text(aes(label= Weights),
            position=position_dodge(width=0.9),
            vjust=-0.25)
```

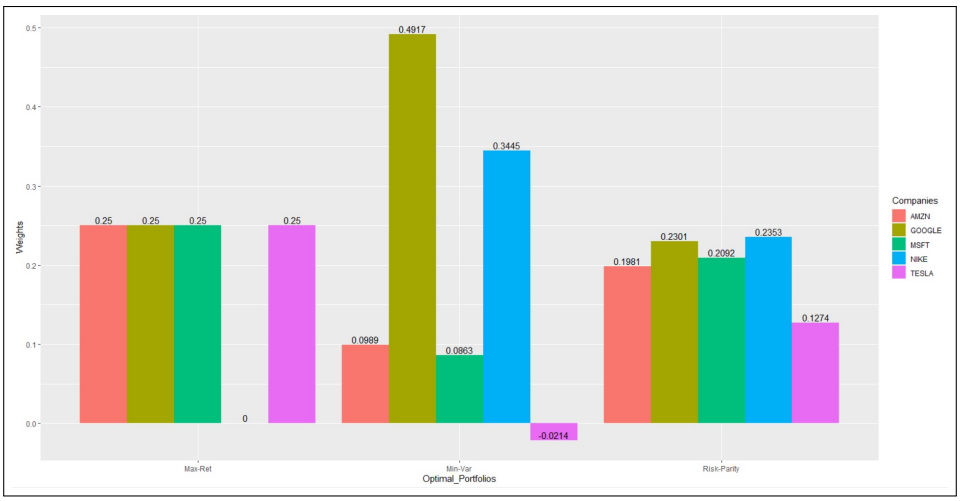

Figure 8: Optimal Portfolio Weights (Comparison).

```r
# Descriptive Statistics [Optimal Portfolios]
library(pastecs)
```

```
##
## Attaching package: 'pastecs'

## The following objects are masked from 'package:xts':
##
##     first, last
```

```r
MinVar = OptPort1
MaxRet = OptPort2
RiskPar = OptPort3

OptPortfolios = cbind(MinVar,MaxRet,RiskPar)

stat.desc(OptPortfolios)
```

```
##                     MinVar        MaxRet       RiskPar
## nbr.val       7.560000e+02  7.560000e+02  7.560000e+02
## nbr.null      0.000000e+00  0.000000e+00  0.000000e+00
## nbr.na        0.000000e+00  0.000000e+00  0.000000e+00
## min          -2.957777e-02 -3.197828e-02 -2.714237e-02
```

```
## max           2.054574e-02  3.946806e-02  2.865303e-02
## range         5.012351e-02  7.144635e-02  5.579540e-02
## sum          -1.967119e+00 -2.368528e+00 -2.082539e+00
## median       -2.490672e-03 -2.773964e-03 -2.451547e-03
## mean         -2.602009e-03 -3.132973e-03 -2.754682e-03
## SE.mean       2.292295e-04  3.261137e-04  2.458924e-04
## CI.mean.0.95  4.500030e-04  6.401973e-04  4.827141e-04
## var           3.972491e-05  8.040069e-05  4.571008e-05
## std.dev       6.302770e-03  8.966643e-03  6.760923e-03
## coef.var     -2.422271e+00 -2.862024e+00 -2.454339e+00
```

```r
# Plotting Optimal Portfolios
windows(8,10)
plot(Date, MinVar, type="l", col="red",
     main="Optimal Portfolio Log Returns (Jan 2020 - Dec 2022)",
     ylab="Return",ylim=c(-0.035,0.04))
lines(Date, MaxRet, col="blue", lty=2)
lines(Date, RiskPar, col="green", lty=2)
grid(nx=20,ny=20)
# Add a legend
legend("topleft", legend=c("MinVar", "MaxRet","RiskPar"),
       col=c("red", "blue","green"),
       lty=1, cex=0.8)
```

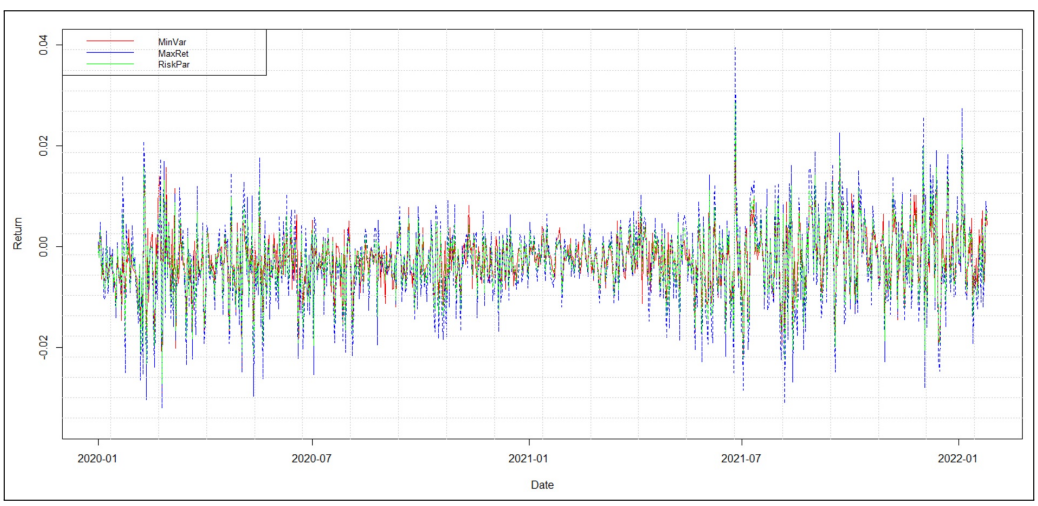

Figure 9: Optimal Portfolio Returns.

```r
# Kernel Density Plot
windows(8,10)
plot(density(MinVar), col = "red",
     main="Densities (Optimal Portfolio Returns)",
     lwd=1.5)
lines(density(MaxRet), col = "blue",lwd=1.5)
lines(density(RiskPar), col = "green",lwd=1.5)

legend("topright",                                  # Add legend to density
       legend = c("Min-Var", "Max-Ret", "Risk-Parity"),
       col = c("red", "blue", "green"),
```

```
       lwd=1.5, lty = 1)
```

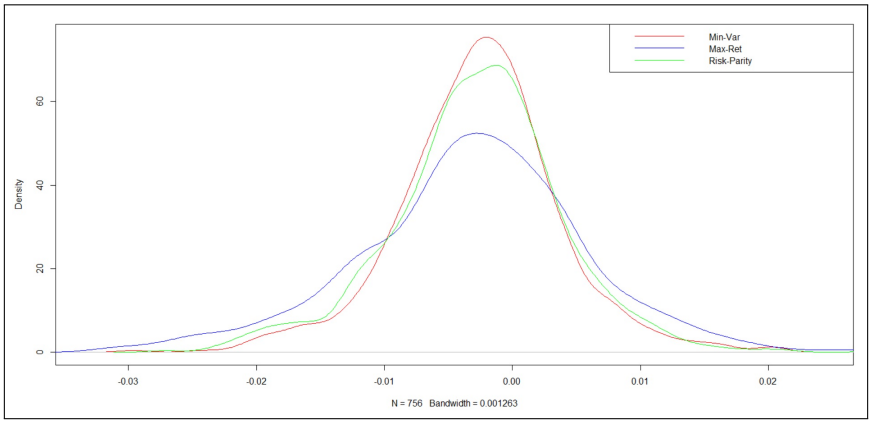

Figure 10: Kernel Density (Optimal Portfolio Returns).

```
# Kernel Density Plot [Polygon]
windows(8,10)
plot(density(MinVar), col = "red",
     main="Densities (Optimal Portfolio Returns)",
     lwd=1.5)
polygon(density(MinVar), col = "red")
lines(density(MaxRet), col = "blue",lwd=1.5)
polygon(density(MaxRet), col = "blue")
lines(density(RiskPar), col = "green",lwd=1.5)
polygon(density(RiskPar), col = "green")
#text(-0.02,60,"Mickey, I hate You :)")

legend("topright",                                  # Add legend to density
       legend = c("Min-Var", "Max-Ret", "Risk-Parity"),
       col = c("red", "blue", "green"),
       lwd=1.5, lty = 1)
```

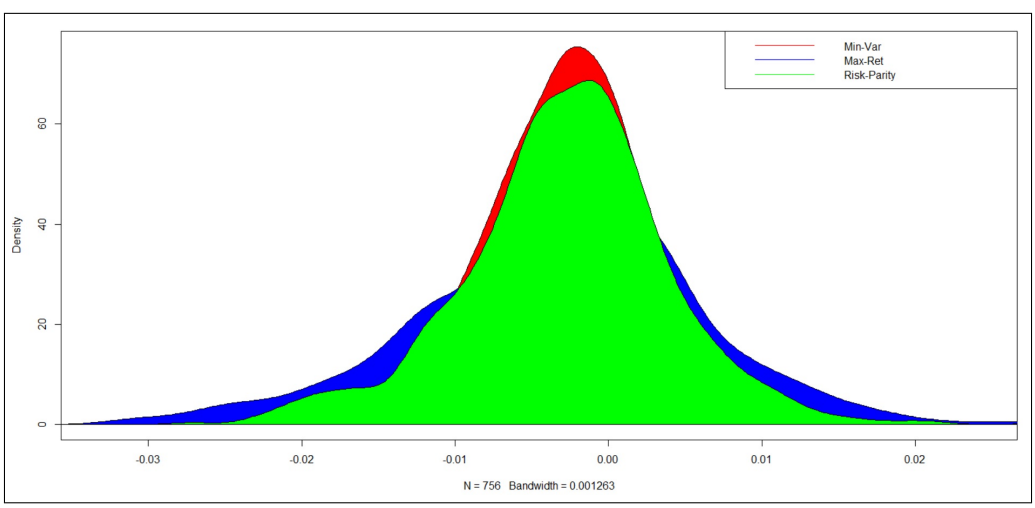

Figure 11: Kernel Density (Optimal Portfolio Returns).

```
#################
# END OF LECTURE.
#################
```

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Northeastern University, Finance Academic Group.

E-Mail: *m.davidovic@northeastern.edu*

Spring 2026.